

NATIONAL OCCUPATIONAL STANDARD FOR SAFETY SPECIALIST



**NOS.SS.01
FIRST EDITION**

APPROVING AUTHORITY

This National Occupational Standard (NOS) has been prepared and published under the authority of the Zambia Qualifications Authority Board on 20th December 2023.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority (ZAQA) Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide for the registration and accreditation of qualifications; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through the various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards (NOS) shall be revised every 5 years, or whenever it is deemed necessary, by the issuance of either amendments or of revised editions. It is important that users of the NOS ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Construction NOSDT, upon which the following organisations were represented:

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2. Engineering Institution of Zambia (EIZ)
3. Ministry of Local Government and Rural Development (MLGRD)
4. Road Development Agency (RDA)
5. Surveyors Institute of Zambia (SIZ)
6. University of Zambia (UNZA)
7. Zambia Institute of Architects
8. Association of Building and Civil Engineering Contractors (ABCEC)
9. Water Resources Management Authority (WARMA)
10. Bari Zambia Limited (BZL)
11. Zulu Barrow Construction (ZBC)
12. Ng'andu Consulting (NC)
13. Zambia Qualifications Authority (ZAQA) – Secretariat
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TABLE OF CONTENTS

FOREWORD	iv
JUSTIFICATION	iv
ACRONYMS AND ABBREVIATIONS	v
GLOSSARY OF TERMS	vi
1. OVERVIEW	1
2. SCOPE	3
3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)	3
4. UNITS AND ELEMENTS	3
5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS	20
6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES	20
7. WORKING CONDITIONS/ENVIRONMENT	20
8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE	21
9. PHYSICAL DEMANDS ON THE BODY	21
ANNEX A	22
Criteria for Assessments based on this NOS	22
ANNEX B	23
NOS Version Control	23

FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“develop and implement a national qualifications framework; register and accredit qualifications; and ensure that standards and registered qualifications are internationally comparable.”***

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through the various sector specific National Occupational Standards Development Teams (NOSDTs) with experts composed of representation from the appropriate authorities, government departments, consumer associations, regulators, industry, academia, and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Construction National Occupational Standards Development Team in accordance with the laid down procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as the National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies, and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula, and learning programmes in various sectors where the occupation exists. In the Construction sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

Safety Specialist is one of the critical trades in the construction industry, which is common to all types of constructions. A Safety Specialist should understand health and safety issues related to building, civil engineering, construction technology processes, construction plant and equipment, and legislation relating to Occupational Health and Safety. A Safety Specialist is important in the industry because he/she ensures that construction safety standards are implemented and adhered to on construction sites, and contributes to reducing occupational health incidences. The development of this National Occupational Standard will ensure relevance of the training to latest advancements in industry, resulting in adequately and appropriately skilled Safety Specialists. This NOS highlights the core knowledge, skills, competences, and personal attributes that Safety Specialists must possess to be successful in their jobs.

ACRONYMS AND ABBREVIATIONS

CS	Core Skills
K	Knowledge and Understanding
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Context
OSHE	Occupational Safety, Health and Environment
PC	Performance Criteria
PS	Professional Skills
RK	Regulatory Context
RPL	Recognition of Prior Learning
S	Skills
SS	Safety Specialist
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Disability: is the physical or mental impairment that substantially limits one or more major life activities.

Function: is an activity necessary for achieving the key purpose of the sector, occupation or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of the NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional, and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy, whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

Elements: set out competences the incumbent should possess to carry out the day-to-day activities.

1. OVERVIEW

This is an introductory section providing a summary and specific information or commentary about the content of the NOS and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.SS.01
Occupation	Safety
Job Title	Safety Specialist
Job Description	A Safety Specialist is responsible for monitoring and assessing hazardous and unsafe situations, and developing measures to assure personnel safety. The Safety Specialist is also responsible for supporting the development and implementation of good safety practices. Further, the Safety Specialist is responsible for supporting the implementation of projects and operations in line with policies, common system standards, and international best practices. As part of his/her duty of care, the Safety Specialist should commit to ensuring that people do not fall sick or sustain injuries when carrying out operations. He/she must ensure a zero-tolerance approach to poor health and safety performance, and support development of health and safety training, and monitoring and compliance. The Safety Specialist also needs to act as Rescue/Emergency Operator.
Job Purpose	A Safety Specialist: • Responds to emergency situations involving hazardous materials, fire and smoke incidents, or unknown hazards; prepares reports of findings and submits recommendations. Further, he/she analyses accident and incident reports for trends; makes recommendations for correcting unsafe conditions and actions. • Assists the Principal Agent, Engineer or Project Manager to effectively implement health and safety management systems. • Reviews practices on construction sites, risk assessments, inspections, incident investigations, and other reports, and offer advice to on-site teams to ensure that good practices are upheld. • Assists site teams to correct deviations from health and safety requirements, including managing breaches of requirements and poor performance by personnel and contractors. • Monitors performance and addresses incidents.

	- Carries out inspections and system reviews to ensure that project operations comply with health and safety, and legal and institutional requirements. - Carries out incident investigations. - Follows-up on actions to address non-conformities or issues raised in incident investigations, inspections, and audits to ensure that the issues have been effectively addressed, and lessons learned shared. - Carries out actions to improve the effectiveness of health and safety critical controls for the project. - Uses trend analysis to monitor data from sites for the purposes of addressing poor performance.
ZQF Level	6
Sector	Construction
Sub sector	Real Estate and Infrastructure Construction
Other Economic Sector(s) in which the Occupation is Practiced	Construction, Manufacturing, Telecommunication, Energy, Education/Training, and other related sectors.
Other Similar Jobs that can be Performed in the Occupation	Safety Coordinator, Safety Superintendent, Safety Supervisor, Health and Safety Manager, etc.
Minimum Educational Job Entry Qualification(s)	Certificate/Diploma in Construction, Engineering
Practicing License Requirements (if any)	No. (But Membership with the Engineering Institution of Zambia is recommended)
Training/RPL (Suggested)	- Training in Occupational Health and Safety in Construction - Diploma or Certification
Minimum Job Entry Age	20 years
Prior Experience (Recommended)	- Non-trained worker: 3 years site experience in the same occupation - Trained worker: 1 year site experience as an apprentice Safety Specialist
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This NOS highlights the core competences, skills, knowledge and understanding, and personal attributes that Safety Specialists are required to possess to be successful in their jobs.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual who has the ability to perform basic planning and prioritising; ability to work in a team, physique to sustain strenuous conditions, high technological exposure to handle various equipment, tools and materials, sensitivity towards safety for self, others, and equipment; willingness to work across various locations in ongoing construction work environments, whilst working at the site; be well-versed with tasks, functions, standards, specifications, codes of practice and safety norms applicable to construction works; be honest and results oriented, etc.

The jobholder develops and implements sustainable business strategies, thinks long-term and externally in order to positively shape the project and/or workplaces; anticipates and perceives the impact and implications of future decisions and activities on other parts of the project and or workplaces; treats all individuals with respect, responds sensitively to differences, and encourages others to do the same; upholds organisational and ethical norms.

The individual must maintain high standards of trustworthiness, must be a role model for diversity and inclusion; evaluates data and courses of action to reach logical and pragmatic decisions; assumes unbiased, rational approach with calculated risks; must apply innovation and creativity to problem-solving. The individual must also express ideas or facts in a clear, concise, and open manner. Effective communication indicates a consideration for the feelings and needs of others; must actively listen and proactively share knowledge, and handle conflict effectively by overcoming differences of opinion and finding common ground.

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 5 units representing the tasks that a jobholder should undertake in his/her day-to-day work. The unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This Unit covers the skills and knowledge required by a Safety Specialist for hazard identification and risk assessment]

Unit No.	01
Unit Title	Hazard Identifications and Risk Assessment (HIRA)
Description	This Unit describes the steps required by a Safety Specialist when working in a construction environment to keep workers and the workplace risk free, and to promote a preventive safety and health culture.
Scope	This Unit covers the following: Steps of Hazard Identification and Risk Assessment, Including Risk Ranking, Risk Control, and Risk Review.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Steps of Hazard Identification and Risk Assessment, Including Risk Ranking, Risk Control, and Risk Review	To be competent, the individual must be able to: PC1. Identify hazards at an early stage of any operation to apply effective measures to prevent accidents. PC2. Conduct hazard identification and risk assessment in accordance with industry standards. PC3. Determine all possible factors that may harm the workers PC4. Assess the risks by determining the probability of accidents and analysing their severity. PC5. Assess the safety risks and ensure they are within acceptable limits. PC6. Identify the existing safeguards and controls. PC7. Make suggestions to counter the likelihood of hazards. PC8. Review the performance of the safeguards and controls.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Workplace safety, health, and wellness policies. OK2. Construction management processes, including procurement, stores, and contract administration. OK3. Disciplinary code of conduct. OK4. Emergency preparedness plans. OK5. Health surveillance schedules.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Baseline risk assessments. TK2. Issue-based risk assessments. TK3. Continuous risk assessments which form part of all inspections and observations that occur on routine basis. TK4. Identification of hazards through field visits and study of all procedures and input documents. TK5. Risk assessment methodology providing a quantitative assessment of potential risks consistent with the planned risk acceptance criterion. TK6. Risk management using appropriate hierarchy of controls TK7. Review of control measure for continuous improvement.

C. Regulatory Context (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. National Council for Construction Act, 2022. RK2. Applicable safety, health, and wellness legislations. RK3. National Occupational Safety and Health Policy.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the site. CS2. Prepare and provide clear and simple instructions, details, and sketches to co-workers. CS3. Prepare reports.
	Reading Skills
	The individual on the job must be able to: CS4. Read English, and be able to or have the means to give simple instructions in the local language used at the site. CS5. Read and interpret sketches, drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS6. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS7. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Determine and report to the superior whether the working space is safe for operations.
	Plan and Organise
	The individual on the job should be able to: PS2. Plan work and organise required resources in coordination with team members and superiors.
	Customer Centricity
	The individual on the job should be able to: PS3. Meet the safety requirements of the client and well-being of all employees.
	Problem-Solving
	The individual on the job should be able to: PS4. Resolve any conflicts within the team.
	Analytical Thinking
	The individual on the job should be able to: PS5. Analyse and convey to the superior and carry out remedial action.

	Critical Thinking
	The individual on the job should be able to: PS6. Identify and deal with or report violation of any safety norms which may lead to accidents.

UNIT 2 [This Unit covers the skills and knowledge required by a Safety Specialist in relations to safety, health, and wellness]

Unit No.	02
Unit Title	Safety, Health, and Wellness
Description	This Unit describes the skills and knowledge required by a Safety Specialist to reduce work related Injuries and accidents.
Scope	This unit covers the following: • Safety, Health, and Wellness Activities. • Emergency Management. • Mental and Emotional Well-being Practices. • Conduct Training on Safety Rules and Regulations; Safe Equipment Usage.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Safety, Health, and Wellness Activities	To be competent, the individual must be able to: PC1. Develop accident prevention and wellness programmes to cover all levels of employees. PC2. Prepare a safety, health, and wellness plan. PC3. Monitor safety, health, and wellness measures. PC4. Cultivate a safety, health, and wellness culture. PC5. Prepare safety, health, and wellness reports.
Emergency Management	To be competent, the individual must be able to: PC6. Create an emergency action plan (EAP). PC7. Perform actions to avoid emergencies, including deterrence and surveillance. PC8. Execute actions in the EAP in the event of emergencies. PC9. Document and report emergencies. PC10. Plan and execute emergency drills to ensure readiness of the work force.
Mental and Emotional Well-being Practices	To be competent, the individual must be able to: PC11. Provide mental and emotional health support. PC12. Monitor work conditions to track mental and emotional health triggers, including stress, anxiety, depression, and fatigue. PC13. Check, monitor, and mitigate against substance abuse. PC14. Create a mentally healthy organisation culture. PC15. Conduct mental and emotional well-being awareness training.
Conduct Training on Safety Rules and Regulations; Safe Equipment Usage	To be competent, the individual must be able to: PC16. Carry out safety awareness and induction of newly recruited workers. PC17. Plan and organise refresher training courses on safety for existing workers. PC18. Create awareness on safety equipment for new workers. PC19. Plan and organise refresher training courses on safety equipment for existing workers. PC20. Organise periodic meetings for the safety team/section.

	PC21. Support personnel in charge of vocational training in conducting regular trainings to ensure safety norms are communicated and understood. PC22. Carry out accident/incident investigations.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of the: OK1. Relevant safety, health, and wellness standards, policies, and procedures followed in the company. OK2. Different types of safety requirements at the construction site. OK3. Procurement, store and inventory management, quality management, and key contact points for query resolution processes. OK4. Organisation safety culture.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Hazard perception, identification, and monitoring. TK2. Accident prevention. TK3. Risk management. TK4. Preparation of safer and health plans. TK5. Preparation of emergency action plans. TK6. Preparation of safety reports. TK7. Standards of relevant statutory bodies and compliance procedures within the sector. TK8. Construction safety standards, including noise levels, pollutants, etc. TK9. Relevant personal protective equipment. TK10. Construction technology techniques. TK11. Construction management processes, including procurement and contract administration. TK12. Communication skills, interpretation of drawings, and contract documentations.
C. Regulatory context (Knowledge of construction safety rules and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. National and international standards for accident prevention. RK2. Occupational Health and Safety management (ISO 45001). RK3. Different types of constructions and details of the construction site being worked in. RK4. Construction organisation, time keeping, need for discipline, and punctuality. RK5. Benching in quarries/open cast constructions, dressing of overhangs, undercuts, fencing, First Aid, and hygiene. RK6. Standing orders in force at the construction site, and safety in the vicinity of the machinery. RK7. Shot firing and safety regulations; how and where to take shelter.

	RK8. Duties of workers/helpers. RK9. Working hours and workplace injury compensation as per the relevant regulations. RK10. Construction safety procedures. RK11. Consequences of violating safety procedures.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills The individual on the job must be able to: CS1. Write in English (at least working level) and/or be able to have the means to give simple instructions in the local language used at the site. CS2. Prepare and provide clear and simple instructions, details, and sketches to co-workers. CS3. Prepare and write instructions, guidelines, sign boards, safety rules, and safety tags for use on site. CS4. Note down observations (if any). CS5. Fill in documentation or enter information in online systems under the guidance of the supervisor.
	Reading Skills The individual on the job must be able to: CS6. Read English, and be able to or have the means to give simple instructions in the local language used at the site. CS7. Read and interpret sketches, drawings or instructions provided for the required work. CS8. Read instructions, guidelines, sign boards, safety rules, and safety tags. CS9. Read instructions and identify exit routes during emergencies.
	Oral Communication (Listening and Speaking Skills) The individual on the job must be able to: CS10. Effectively communicate with superiors, colleagues, and regulators. CS11. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS12. Convey information clearly and concisely to co-workers.
	Decision-Making The individual on the job must be able to: PS1. Determine whether the working space is safe for operations. PS2. Determine whether safety measures in place are adequate for the level of potential risks and hazards. PS3. Make decisions in emergency situations.
B. Professional Skills	Plan and Organise The individual on the job should be able to: PS4. Plan work and organise required resources in coordination with team members to ensure adherence to safety standards.

	Customer Centricity
	The individual on the job should be able to: PS5. Focus on meeting the safety requirements of the client and well-being of all employees.
	Problem-Solving
	The individual on the job should be able to: PS6. Resolve any conflicts and problems related to safety, health, and well-being of workers. PS7. Detect problems in day-to-day tasks. PS8. Discuss possible solutions to address problems, with the supervisor.
	Analytical Thinking
	The individual on the job should be able to: PS9. Analyse and convey to the superior and carry out remedial action on any violations of safety norms.
	Critical Thinking
	The individual on the job should be able to: PS10. Identify and deal with and report violations of any safety norms which may lead to accidents.

UNIT 3 [This Unit covers the skills and knowledge required by a Safety Specialist in maintaining safety and health measures when working with dangerous and hazardous materials critical in construction]

Unit No.	03
Unit Title	Maintaining Health and Safety Measures
Description	This Unit describes the skills and knowledge required by a Safety Specialist in maintaining safety and health measures critical on construction sites when using dangerous and hazardous materials.
Scope	This Unit covers the following: • Maintaining Safety and Health Measures Critical in Construction.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Maintaining Safety and Health Measures Critical in Construction	To be competent, the individual must be able to: PC1. Comply with occupational health and safety regulations in the industry. PC2. Adhere to construction technology and operating procedures with respect to workplace health and safety, and with dangerous and hazardous materials. PC3. Follow the correct safety steps in case of accidents, fire or major failure. PC4. Comply with safety regulations and procedures in case of fire hazards. PC5. Operate various types of fire extinguishers. PC6. Work responsibly, safely, and as carefully as possible to avoid endangering self or others, including members of the public and equipment. PC7. Perform storage and transportation of hazardous materials in compliance with safety guidelines. PC8. Deal with misfires as per statutory requirements. PC9. Identify characteristics of post-blast fumes and take necessary precautions. PC10. Wear safety gear, such as hardhat, respiratory protection, eye protection, ear protection, and maintaining safe distances. PC11. Adhere to the manufacturer's instructions for care and safe operation of the equipment.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Relevant standards and procedures followed in the industry when dealing with dangerous and hazardous materials. OK2. Different types of safety requirements at the construction site.

	OK3. Processes such as procurement, stores management, inventory management, quality management, and key contact points for query resolution.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Construction safety rules and regulations. TK2. Construction safety standards, including noise levels, pollutants, etc. TK3. Relevant personal protective equipment. TK4. Primary tier of occupational safety and health (accident prevention, standard operating procedures, etc.), secondary tier (rescue operations, periodical medical exams, and First Aid provision), and tertiary tier (counselling and rehabilitation, and compensation).
C. Regulatory Context (Knowledge of construction safety rules and regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Benching in quarries, dressing of overhangs, undercuts, and fencing. RK2. First Aid and hygiene. RK3. Code of traffic in specific areas of the construction site and significance of fences. RK4. Standing orders in force at the construction site and safety in the vicinity of the machinery. RK5. Shot firing and safety regulations, and how and where to take shelter. RK6. Construction safety procedures. RK7. Consequences of violating safety procedures. RK8. In-House emergency preparedness or disaster management plan. RK9. Environmental impact of construction. RK10. Sources of dust, noise and vibration, and measures to minimise them. RK11. Hazardous material safety, and security rules and regulations as prescribed by the Constructions Safety department. RK12. Code of practice for safe handling and transportation of dangerous material and heavy equipment.
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and interpret symbols and measurements. CS2. Read information documents. CS3. Analyse the available data about the site.
	Writing Skills
	The individual on the job must be able to: CS4. Note down observations (if any). CS5. Fill in documentation or enter information in online systems under the guidance of the supervisor.

	Oral Communication (Listening and Speaking Skills) The individual on the job must be able to: CS6. Discuss task lists, schedules, and activities. CS7. Effectively communicate with superiors, colleagues, and regulators. CS8. Listen attentively and comprehend the information given by various sources about the site.
	Decision-Making The individual on the job must be able to: PS1. Determine and report to the superior whether the work space is safe for operations. PS2. Detect problems in day-to-day tasks. PS3. Discuss possible solutions to address problems with the supervisor. PS4. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation).
B. Professional Skills	Plan and Organise The individual on the job must be able to: PS5. Plan and organise the work order and jobs. PS6. Organise all process manuals so that sorting out or accessing information is easy.
	Customer Centricity The individual on the job should be able to: PS7. Focus on meeting the safety requirements of the client and well-being of all employees.
	Problem-Solving The individual on the job should be able to: PS8. Resolve any conflicts within the team.
	Analytical Thinking The individual on the job should be able to: PS9. Analyse information and convey to the superior and carry out remedial action.
	Critical Thinking The individual on the job should be able to: PS10. Identify and deal with or report violation of any safety norms which may lead to accidents.

UNIT 4 [This Unit covers the skills and knowledge required by a Safety Specialist in working with dangerous and hazardous materials]

Unit No.	04
Unit Title	Working with Dangerous and Hazardous Materials
Description	This Unit describes the skills and knowledge required by a Safety Specialist in working with dangerous and hazardous materials.
Scope	This Unit covers the following; Formal Top-down Organisation-wide Approach to Managing Safety Risk; Assuring Effectiveness of Controls in Construction Sites.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Formal Top-down Organisation-wide Approach to Managing Safety Risk; Assuring Effectiveness of Controls in Construction Sites	To be competent, the individual must be able to: PC1. Develop and enforce occupational safety policy and objectives. PC2. Apply construction safety risk management. PC3. Prepare safety assurance construction during operations PC4. Promote safety culture as a project and company's core value. PC5. Demonstrate knowledge transfer to employees.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Safety as an integral part of company values, demonstrating commitment daily. OK2. Anticipating and mitigating risks at the process level system description and task analysis; hazard identification, risk analysis, risk assessment, and risk control in construction. OK3. Common safety assurance functions include: internal audits, investigations, and employee reporting systems. OK4. Safety management systems, and learn from shared experience and develop awareness of hazards.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Construction safety legal and institutional framework. TK2. Construction safety standards and corporate due diligence. TK3. Quality management system (QMS) aligned with organisational safety, health, and environmental (OHSE) management systems. TK4. Organisation-wide approach to managing safety risk, and assuring the effectiveness of controls.
C. Regulatory Context (Knowledge of constructions)	The individual on the job must demonstrate knowledge and understanding of: RK1. Compliance strategy.

safety rules and regulations)	RK2. Industry specific regulations and legal system. RK3. Codes of practice. RK4. Standing orders in force in the construction. RK5. Construction safety procedures. RK6. Enforcement and consequences of violating safety procedures.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Note down observations (if any). CS2. Fill in documentation or enter information in online systems.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret strategic management documents. CS4. Read and interpret legal documents. CS5. Understand and analyse the available data about the trends in the industry's safety management
	Oral Communication (Listening and Speaking Skills)
The individual on the job must be able to: CS6. Discuss task lists, schedules, and activities. CS7. Effectively communicate with superiors, colleagues, and regulators. CS8. Listen attentively and comprehend the information given by various sources about the project,	
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Detect problems in day-to-day tasks. PS2. Discuss possible solutions to address problems. PS3. Make decisions in emergency situations, as per the authority matrix defined by the organisation.
	Plan and Organise
	The individual on the job must be able to: PS4. Plan and organise the work order and jobs. PS5. Organise all process manuals so that sorting out or accessing information is easy.
	Customer Centricity
The individual on the job should be able to: PS6. Meet the safety requirements of the client and well-being of all employees.	
Problem-Solving	
The individual on the job should be able to: PS7. Resolve any conflicts within the team.	

	Analytical Thinking
	The individual on the job should be able to:
	PS8. Analyse and convey to the superior and carry out remedial action.
	PS9. Use reasoning skills to identify and resolve basic problems.
	PS10. Detect any potential problems which could arise.
	Critical Thinking
	The individual on the job should be able to:
	PS11. Identify and deal with or report violation of any safety norms which may lead to accidents.
	PS12. Follow instructions and work on areas of improvement identified from regulatory and enforcement bodies.
	PS13. Handle safety and health tasks with minimum supervision.
	PS14. Complete the job within agreed timelines and quality norms.

UNIT 5 [This Unit covers the skills and knowledge required by a Safety Specialist in compliance with applicable safety and health legislation]

Unit No.	05
Unit Title	Compliance with Applicable Legislation
Description	This Unit describes the skills and knowledge required by a Safety Specialist to demonstrate awareness of applicable safety and health legislation.
Scope	This Unit covers the following: Compliance with Applicable Safety and Health Legislation.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Compliance with Applicable Safety and Health Legislation	To be competent, the individual must be able to: PC1. Demonstrate knowledge in the understanding of the Occupational Safety and Health Act, Factories Act, Environmental Management Act, The Explosives Act, and other relevant legislation. PC2. Maintain relevant documentation required under various legislation. PC3. Maintain organisational and statutory reports regarding inspections, testing of plant and equipment, and incident investigations. PC4. Maintain records of the maintenance, servicing, and testing done for emergency preparedness in facilities. PC5. Demonstrate ability to apply relevant codes of practice.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Relevant standards and procedures followed in the company. OK2. Different types of safety requirements on site. OK3. Internal processes and management systems. OK4. Coordination mechanisms with stakeholders/contractors.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Occupational safety and health auditing. TK2. Site inspections and incident investigations. TK3. Developing and implementing safe working procedures. TK4. Communication mechanisms with stakeholders.
C. Regulatory Context (Knowledge of relevant applicable legislation)	The individual on the job must demonstrate knowledge and understanding of the: RK1. National Occupational Safety, Health and Wellness Policy. RK2. Organisational Safety, Health and Wellness Policy. RK3. Provisions of the National Council for Construction Act, 2020. RK4. Provisions of all national occupational safety, health, wellness, and environmental management legislation.

	RK5. Global standards on occupational safety and health management.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to:
	CS1. Write in English (at least working level) and/or be able to have the means to give simple instructions in the local language used at the site.
	CS2. Prepare and provide clear and simple instructions, details, and sketches to co-workers.
	CS3. Generate reports.
	Reading Skills
	The individual on the job must be able to:
	CS4. Read English, and be able to or have the means to give simple instructions in the local language used at the site.
	CS5. Read and interpret sketches, drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to:
	CS6. Listen attentively and interpret communication/ instructions from the supervisor and co-workers.
	CS7. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to:
	PS1. Determine and report to the superior aspects of the work site which may not be in conformity with the legal requirements.
	Plan and Organise
	The individual on the job should be able to:
	PS2. Plan work and organise required resources to ensure conformity with all statutory requirements.
	Customer Centricity
	The individual on the job should be able to:
	PS3. Meet the safety requirements of the client and well-being of all employees.
	Problem-Solving
	The individual on the job should be able to:
	PS4. Apply HIRA to determine and implement control measures to ensure conformity to statutory requirements.
	Analytical Thinking
	The individual on the job should be able to:
	PS5. Analyse the effectiveness of control measures for continuous improvement.
	PS6. Analyse near misses in order to prevent potential accidents.

	Critical Thinking
	The individual on the job should be able to: PS7. Identify and deal with or report violation of any safety norms which may lead to accidents.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

Equipment, tools, and consumable materials used by the jobholder include, but are not limited to:

- Hand held alcohol breathalyser;
- Pocket size First Aid kit;
- Sound level meter;
- Measuring tape.

6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES

Dilemmas associated with the job of Safety Specialist include: exposure to flames and electrical power supply; working around and with machinery with moving parts; working in dangerous areas with likelihood of sharp or falling materials and objects; working in confined spaces and at heights with likelihood of falls; working in extreme weather such as hot, cold or wet conditions; working in noisy or dusty environments; exposure to welding fumes and odours; lifting/pulling/pushing heavy materials; long working hours; pressure from supervisors, work colleagues, government regulators, etc.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include: wearing protective clothing and ensuring their availability and use by other employees; exercising regularly to maintain physical fitness; exercising proper work ergonomics; participating in workplace safety sensitisation and awareness meetings/training sessions; adhering to company's safety and standard operating procedures at all times; consulting extensively within and outside one's department/team on construction safety issues; planning and prioritising work, etc.

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include: indoor and outdoor construction sites, as well as workshops and workplaces; may also work in commercial buildings or private homes, and confined spaces; handling machines with moving parts; working at great heights; working in conditions that may be dirty and noisy; exposure to seasonal heat and cold or adverse weather conditions; emergency call-outs; standing or squatting for long hours, and lifting relatively heavy objects. In most cases, the job involves working normal hours, but in some instances, shift work and regular overtime may be required. The job also requires wearing suitable protective clothing such as works suits, ear protectors, safety visors or goggles, gloves and hardhats, safety harnesses, etc.

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Parties involved or interacting with the jobholder who are internal to the organisation include: supervisors/superiors, trainers, occupational safety and health team, other colleagues, etc.

8.2 External/Outside the Organisation

Parties involved or interacting with the jobholder who are external to the organisation include: government regulators, trainers, clients, suppliers of equipment/tools/consumables, fellow Safety Specialists from other companies and labour unions/occupational safety and health associations, etc.

9. PHYSICAL DEMANDS ON THE BODY

- Ability to sustain strenuous conditions such as climbing heights;
- Walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Lift, carry, push and pull heavy objects;
- Use fingers, hands and feet with ease to complete the assigned task (dexterity);
- Strenuous works that may cause musculoskeletal disorders;
- Etc.

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programme developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programme developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'Out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Individual awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part, and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.SS.01		
ZQF Level	6	Version Number	01
Sector	Construction	Date of Approval	December 2023
Sub Sector	Real Estate and Infrastructure Construction	Date of Last Review	N/A
Occupation	Safety Specialist	Date of Next Review	December 2028



*“You learn,
We Standardise”*

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